

Enterprise Platform Documentation & Technical Manual

Comprehensive Architectural Blueprint, Feature Manual, Security Framework, Logistics Guide & System Administration Handover

PLATFORM OVERVIEW: The Saveur is an enterprise-grade direct-to-consumer (D2C) e-commerce system built with Python Flask, SQLite/PostgreSQL, Redis caching, Celery task queuing, and multi-gateway payment processing. This complete reference document details all client-facing features, administrative controls, logistics lifecycle tracking, automated tax invoicing, security architectures, and operations protocols.

Project Name:	The Saveur E-Commerce Web Platform	Document Version:	v2.4 (Production Release)
Client / Owner:	The Saveur Gourmet Foods	Release Date:	September 2026
Primary Framework:	Python 3.12 / Flask 3.1	Security Model:	HMAC-SHA256 / Alphanumeric Refs
Database Engine:	SQLite 3 (WAL Mode) / PostgreSQL	Logistics Engine:	5-Stage Dynamic Multi-Courier
Payment Gateways:	Razorpay (UPI/Cards) + PayPal + COD	Task Queuing:	Celery + RabbitMQ / Redis

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1. Executive Summary & Core Platform Capabilities

The Saveur web platform is a custom-engineered, full-featured direct-to-consumer (D2C) e-commerce application designed to deliver an opulent shopping experience for artisanal gourmet snacks, roasted makhanas, cookies, and premium dry fruits. The application seamlessly bridges client engagement, ultra-fast browsing, multi-gateway checkout, automated logistics dispatch, and strict data security.

Key Core Capabilities & Recent Enhancements:

- **High-Speed Responsive Storefront:** Clean, modern aesthetic with rich typography, mobile-first responsive layouts, interactive hero carousels, dynamic product filters, and live search.
- **5-Stage Real-Time Logistics Lifecycle:** Complete status synchronization across *Order Confirmed* → *Shipped* → *In Transit* → *Out for Delivery* → *Delivered* with dynamic multi-courier tracking links (Blue Dart, Delhivery, DTDC, India Post, etc.).
- **Automated Single-Page Tax Invoicing:** Print-optimized A4 tax receipts strictly formatted to single-page dimensions, containing complete FSSAI No. compliance, GST breakdown, QR verification, and color-highlighted payment methods.
- **Cryptographic Security & Alphanumeric IDs:** Non-guessable order references (TSV-A3X9KZ2Q), customer-specific authorization preventing unauthorized access (IDOR protection), and HMAC-SHA256 direct access tokens for seamless, passwordless email tracking.
- **Professional Transactional Email Communications:** Completely de-emojified, executive-grade email notifications dispatched asynchronously via Celery + RabbitMQ with instant fallback.
- **Omnichannel Payment Highlighting:** Clear, color-coded visual badges for Cash on Delivery (COD), Prepaid (Razorpay / UPI), and PayPal across the Admin Console, Customer Orders, Tracking, Invoices, and Checkout.

2. System Architecture & Technical Specifications

The application is architected around a modular Flask backend utilizing the Application Factory and Blueprint design patterns. This ensures clean separation of concerns, high maintainability, and horizontal scalability.

Layer / Component	Technology Stack	Functional Role & Description
Web Server & Proxy	Nginx / HTTP/2 / SSL	Terminates TLS/SSL, serves static assets (CSS, JS, WebP images) with gzip compression, reverse-proxies requests to Gunicorn.
WSGI Application Server	Gunicorn / Flask 3.1.3	Multi-worker Python WSGI server executing route controllers, Jinja2 template rendering, authentication, and business logic.
In-Memory Cache	Redis 8.0+	High-speed key-value cache storing navigation categories, product catalogs, and state shipping rates with auto-invalidation on updates.
Task Queue & Broker	Celery 5.6 + RabbitMQ	Asynchronous background queue handling SMTP email dispatch (OTPs, order confirmations, shipping updates) without blocking HTTP requests.
Database Engine	SQLite 3 (WAL Mode) / PostgreSQL	ACID-compliant relational database holding users, products, orders, coupons, reviews, and logs. Configured with Write-Ahead Logging.
Payment Gateways	Razorpay REST + PayPal SDK	Cryptographically verified payment gateways supporting UPI, Credit/Debit Cards, Net Banking, Wallets, and International payments.

Key Architectural Advantages:

- Zero-Lag Background Processing:** Email delivery takes place in background worker threads; customers experience instantaneous checkout without waiting for SMTP handshakes.
- Resilient Graceful Fallbacks:** If Redis or RabbitMQ are temporarily unreachable, the application automatically catches exceptions and falls back to direct database queries and synchronous email delivery without user interruption.
- Modular Directory Structure:** Code is organized into dedicated service layers (services/), modular blueprints (routes/), context processors, database managers, and clean Jinja2 template components.

3. Storefront Features & Customer Journey

The customer experience has been engineered for maximum conversion, visual delight, and effortless navigation across all desktop, tablet, and mobile devices.

Journey Stage	Key Features & Capabilities	Customer Benefit
1. Discovery & Browsing	Interactive hero carousel, category grids, subcategory filtering, live search with instant suggestions, bestseller highlight badges.	Effortless product discovery and brand engagement.
2. Product Evaluation	Multi-angle image gallery with zoom, unit weight specs (e.g., 85g Jar), stock availability badges, discount pricing comparison, verified customer reviews.	Clear purchasing context and confidence.
3. Cart & Wishlist	Persistent cart across sessions, one-click quantity adjustments, coupon code validator with real-time discount calculation, state-based shipping calculator.	Transparent pricing with zero checkout surprises.
4. Secure Checkout	Dual-step checkout: Shipping Address & Contact details, followed by Payment Selection (Razorpay UPI/Cards, PayPal, or Cash on Delivery).	Frictionless, flexible payment options for domestic & NRI customers.
5. Post-Purchase	Instant order confirmation page, alphanumeric reference ID, downloadable single-page tax invoice, and live multi-courier tracking link.	Immediate reassurance and transparent delivery visibility.

Highlighting of Payment Methods Across the Storefront:

Every touchpoint in the customer journey clearly identifies the payment mode used with branded visual pill tags:

Payment Method	Visual Pill Styling	Usage & Verification
Cash on Delivery (COD)	[Amber Pill: Cash on Delivery]	Pay upon delivery. Supported across all major postal pin codes.
Razorpay (Prepaid)	[Emerald Pill: Prepaid (Razorpay)]	Instant payment via UPI (GPay, PhonePe, Paytm), Credit/Debit Cards, Net Banking.
PayPal (Prepaid)	[Blue Pill: Prepaid (PayPal)]	International currency payments captured via PayPal REST API.

4. End-to-End Logistics & 5-Stage Live Order Tracking

Logistics transparency is a cornerstone of The Saveur's customer trust. The platform features an automated, multi-courier fulfillment workflow that synchronizes administrative operations with real-time customer tracking.

Sta ge	Status Name	Trigger & Operational Workflow	Customer Visibility & UI State
1	Order Confirmed	Triggered immediately upon successful payment or COD placement. Order received in kitchen/warehouse queue.	Stage 1 marked Completed with green checkmark. Customer receives immediate confirmation email.
2	Shipped	Admin enters Courier Partner & AWB tracking number. System automatically promotes status to Shipped.	Stage 2 marked Completed. Customer receives dispatch email with clickable AWB live tracking link.
3	In Transit	Package is en route between logistics distribution hubs and sorting centers.	Stage 3 marked In Progress / Completed. Stepper updates dynamically.
4	Out for Delivery	Package assigned to last-mile delivery agent for same-day handover.	Stage 4 highlighted. Customer receives 'Out for Delivery' heads-up email alert.
5	Delivered	Package handed over to customer. Admin marks order as Delivered.	All 5 stages marked Completed with green checkmarks. Delivery confirmation email sent.

Supported Courier Partners & Dynamic Tracking:

Blue Dart: <code>bluedart.com/tracking?track={awb}</code>	Delhivery: <code>delhivery.com/track/package/{awb}</code>
DTDC: <code>dtdc.in/tracking/shipment-tracking.asp?trk={awb}</code>	India Post: <code>indiapost.gov.in/_layouts/15/dop.portal.tracking/trackconsignment.aspx</code>
Shadowfax: <code>shadowfax.in/track?awb={awb}</code>	Xpressbees: <code>xpressbees.com/track?isawb=true&trackid;={awb}</code>
Ekart Logistics: <code>ekartlogistics.com/shipmenttrack/{awb}</code>	Ecom Express: <code>ecomexpress.in/tracking/?awb={awb}</code>

5. Automated Single-Page Invoicing & Tax Compliance

The invoicing engine generates compliant, beautiful, single-page A4 tax receipts for every order. It is accessible both from the customer dashboard and the administrative console.

Compliance / Design Standard	Implementation Specification & Detail
Strict Single-Page Guarantee	Engineered with CSS @media print rules, compact margins, and clean flexbox rows ensuring zero page overflow or awkward 2nd page spills.
FSSAI Registration Display	Prominently features standardized 'FSSAI No.' branding, license number, and statutory food safety compliance marks.
GST & Tax Breakdown	Itemizes product pricing, applied promotional discounts, state shipping charges, and inclusive GST calculation breakdown.
Payment Method Highlighting	Displays color-coded badges for payment modes: Cash on Delivery (Amber) , Prepaid Online / Razorpay (Emerald) , PayPal (Blue) .
Cryptographic QR Verification	Includes a dynamic QR code encoding the secure order access URL for rapid mobile scan-to-verify authenticity.

Invoice Layout Structure Breakdown:

- **Header Section:** The Saveur brand logo, Tax Invoice title, Official FSSAI No., GSTIN, and Registered Corporate Address.
- **Metadata Strip:** Order Reference Number (e.g. TSV-A3X9KZ2Q), Invoice Date, Payment Status Badge, and Highlighted Payment Mode Badge.
- **Billed To & Shipped To:** Customer full name, shipping address, city, state, postal PIN code, and contact phone number.
- **Itemized Products Table:** S.No, Item Description, Unit Specification (e.g. 85g Jar), Unit Price, Quantity, and Line Item Total.
- **Financial Summary:** Subtotal, Item-level Discounts, Applied Coupon Code savings, State Shipping & Logistics fee, Inclusive GST, and Final Total.

6. Transactional Email System & Notification Dispatch

Customer communications follow strict enterprise design standards. All informal emojis have been eliminated in favor of clean typography, structured data cards, and secure action buttons.

Email Event	Subject Line Standard	Key Content & Call-to-Action
Order Confirmed	Order Confirmed: {order_number} - The Saveur	Order summary, itemized price breakdown, delivery address, highlighted payment mode, and 'Track Your Order' button.
Order Shipped	Your Order {order_number} Has Been Shipped - The Saveur	Courier Partner name, AWB Tracking number, estimated delivery date, and direct link to carrier tracking.
Out for Delivery	Out for Delivery: Order {order_number} - The Saveur	Same-day delivery heads-up alert, courier contact info, and recipient preparation guidelines.
Order Delivered	Delivered: Order {order_number} - The Saveur	Handover confirmation, single-page tax invoice access link, and customer feedback invitation.
Account Security	Your Verification Code / Password Reset - The Saveur	6-digit cryptographic OTP, 10-minute expiry warning, and unauthorized access advisory.

Email Engineering Standards:

- **100% Emoji-Free Policy:** Clean, high-trust text layout matching Fortune 500 corporate communications.
- **Cryptographic Link Authentication:** All action buttons (e.g. *Track Order*, *Download Invoice*) contain signed HMAC tokens allowing instant, password-free view for the genuine recipient while maintaining zero risk to their account credentials.
- **Asynchronous Task Dispatch:** Celery workers deliver emails in the background; customer orders are confirmed in under 150 milliseconds.

7. Security Framework, Cryptographic Tokens & Authorization

The Saveur implements a multi-layered security architecture protecting customer privacy, payment data, and administrative functionality against common web vulnerabilities.

Security Layer	Mechanism & Standard	Protection Objective
Secure Alphanumeric IDs	Replaced sequential database IDs with TSV-XXXXXXX references.	Prevents competitor order-volume scraping and sequential guessing attacks.
HMAC-SHA256 Tokenization	Cryptographically signed tokens embedded in email direct-links: ?token=...	Allows seamless one-click tracking/invoice viewing from email without exposing login sessions.
Customer Ownership Auth	Strict session-to-order binding (session['user_id'] == order['user_id']).	Completely prevents Insecure Direct Object References (IDOR).
Payment Signature Verification	Razorpay HMAC-SHA256 signature verification & PayPal REST OAuth capture.	Prevents fraudulent checkout injection, spoofed amounts, and unpaid order approvals.
Password & Session Security	PBKDF2 SHA-256 password hashing with salt, HTTPOnly cookies, CSRF defenses.	Guarantees credential safety and prevents session hijacking.

Cryptographic Access Verification Flow:

When an order tracking or invoice URL is requested (e.g. /track-order/TSV-A3X9KZ2Q):

1. If an Admin session is active, full operational access is granted immediately.
2. If a Customer session is active, the system checks if session['user_id'] == order['user_id']. If matched, access is granted; if mismatched, access is blocked.
3. If no session is active (e.g., opened directly from customer email), the system evaluates the ?token= query parameter against the HMAC-SHA256 signature generated with the server secret key. If valid, secure read-only access is granted.

8. Administrative Control Portal & Operations Manual

The administrative panel (/admin) provides business operators with a centralized, intuitive command center to oversee orders, catalog inventory, shipping rates, promotional campaigns, and customer service inquiries.

Module	URL Route	Key Capabilities & Operations
Executive Dashboard	/admin	Live revenue metrics, today's order counts, low-stock alerts, recent customer transactions, and order distribution graphs.
Order Fulfillment	/admin/orders	Search & filter orders by status, view customer profile & delivery coordinates, assign Courier Partner & AWB number, print tax invoices.
Product Catalog	/admin/products	Add/Edit/Delete products, manage unit weights, upload primary & secondary gallery images, adjust stock counts, toggle Bestseller tags.
Categories & Taxonomy	/admin/categories	Create top-level food categories (e.g. Makhana, Dry Fruits, Cookies) and nested subcategories with image banners.
Couriers & Logistics	/admin/couriers	Add new courier carriers, define live tracking URL templates with {awb} placeholders, manage carrier statuses.
Shipping Rates Engine	/admin/shipping	Configure state-specific shipping charges across all 36 Indian States & Union Territories.
Promos & Coupons	/admin/promos	Create percentage or flat discount coupon codes, set minimum order values, define usage limits and expiry dates.
Hero Banner Slides	/admin/slides	Manage homepage carousel imagery, marketing headlines, subheadings, and direct call-to-action button links.
Customer Enquiries	/admin/enquiries	Review customer contact messages, bulk corporate gifting inquiries, and direct reply records.
System Diagnostics	/admin/system	Clear Redis cache, inspect Celery queue depth, monitor database size, view error logs.

Order Fulfillment Guide:

- Open the order in **Admin** → **Orders** → **View Order**.
- Review items and pack package with packing slip.
- Select **Courier Partner** and paste the **AWB / Tracking Number** in the logistics card.
- Click **Save**. The order status automatically transitions to **Shipped** and customer is notified via email with live tracking link.

9. Database Relational Schema & Entity Specifications

The relational database contains 14 structured tables maintaining foreign key integrity and indexing on critical lookup columns (e.g. order_number, email, category).

Table Name	Primary Key	Key Columns & Relationships
users	id (TEXT)	full_name, email (UNIQUE), password_hash, is_admin, phone, shipping_address, city, state, zip_code, created_at
products	id (TEXT)	name, category, sub_category, description, image_filename, price, stocks, is_bestseller, unit, shipping_charge, gst_rate, discount_percent
product_images	id (INTEGER)	product_id (FK → products.id), image_filename, is_primary, display_order
categories	id (INTEGER)	name (UNIQUE), image_filename, display_order
subcategories	id (INTEGER)	category_id (FK → categories.id), name, display_order
orders	id (INTEGER)	user_id (FK → users.id), order_number (UNIQUE), total_amount, shipping_address, city, state, zip_code, payment_method, status, contact_name, contact_email, contact_phone, courier_partner, tracking_number, estimated_delivery, razorpay_order_id, razorpay_payment_id, paypal_order_id, discount_amount, promo_code, shipping_charge, created_at
order_items	id (INTEGER)	order_id (FK → orders.id), product_id (FK → products.id), quantity, price, original_price, discount_percent
location_shipping_charges	id (INTEGER)	state (UNIQUE), charge
couriers	id (INTEGER)	name (UNIQUE), tracking_url_template, is_active, display_order
promo_codes	id (INTEGER)	code (UNIQUE), discount_percent, flat_discount, min_order_amount, max_discount, valid_until, is_active
carousel_slides	id (INTEGER)	title, subtitle, button_text, button_link, image_filename, display_order, is_active
reviews	id (INTEGER)	product_id (FK → products.id), user_id (FK → users.id), rating, review_text, created_at
enquiries	id (INTEGER)	name, email, phone, subject, message, created_at
password_resets	id (INTEGER)	email, otp_hash, expires_at, used

10. API Directory, Routing Architecture & Endpoints

The system exposes a clean RESTful URL hierarchy mapped to modular Flask Blueprints:

Endpoint Pattern	HTTP Method	Blueprint / Handler	Description & Access Level
/	GET	storefront.index	Homepage with hero slides, categories, and bestsellers.
/shop	GET	products.shop	Catalog with price/category/subcategory filters.
/product/<id>	GET	products.product_detail	Product detail view, image gallery, specs & reviews.
/cart	GET/POST	cart.view_cart	Cart management, item updates, promo application.
/checkout	GET/POST	checkout.checkout_step1	Shipping address capture & state shipping calculation.
/checkout/payment	GET/POST	checkout.checkout_step2	Payment selection, Razorpay order creation, PayPal capture.
/track-order/<order_ref>	GET	orders.track_order	Secure live tracking page (HMAC / User auth).
/orders/<order_ref>/invoice	GET	orders.order_invoice	Customer single-page tax invoice print/view.
/my-orders	GET	orders.my_orders	Customer account order history with status pills.
/login, /register, /logout	GET/POST	auth.*	Customer authentication and session management.
/admin/*	GET/POST	admin.*	Admin management suites (Admin role required).

11. Deployment, Environment Variables & Maintenance Guide

Key Environment Configuration Variables (.env):

Variable Name	Default / Example	Purpose & Operational Requirement
SECRET_KEY	hex_string_64_bytes	Cryptographic signing key for sessions, cookies, and HMAC order access tokens.
DATABASE_URL	sqlite:///thesaveur.db	Connection string for SQLite database or remote PostgreSQL cluster.
REDIS_URL	redis://localhost:6379/0	Cache broker URL for taxonomy and catalog acceleration.
CELERY_BROKER_URL	amqp://guest:guest@localhost:5672//	RabbitMQ message broker URL for asynchronous task dispatch.
RAZORPAY_KEY_ID / SECRET	rzp_live_... / sec_...	Live API keys for domestic UPI, cards, and netbanking processing.
PAYPAL_CLIENT_ID / SECRET	client_... / secret_...	REST API credentials for international multi-currency checkout.
MAIL_SERVER / USER / PASSWORD	smtp.hostinger.com:465	SSL/TLS SMTP server credentials for transactional email delivery.

Routine Maintenance & Backup Instructions:

- **Database Backups:** The SQLite database file thesaveur.db should be backed up daily via cron to secure cloud storage (AWS S3 / Google Cloud Storage).
- **Media Asset Storage:** All uploaded product photos and carousel banners are persisted under static/images/products/ and static/images/carousel/.
- **One-Command Production Deployment:** Use ./deploy.sh to perform git pulls, dependency updates, and zero-downtime Gunicorn worker restarts on Linux servers.

12. Client Deliverables Sign-Off & Handover

All requested business requirements, UI improvements, logistics integrations, and security hardening have been implemented, tested, and validated against production standards.

Feature / Deliverable	Scope Delivered	Verification Status
Single-Page Tax Invoicing	Compact CSS print rules, FSSAI No. compliance, GST breakdown, QR verification.	VERIFIED (100%)
5-Stage Logistics Stepper	Order Confirmed → Shipped → In Transit → Out for Delivery → Delivered.	VERIFIED (100%)
Dynamic Multi-Courier Integration	Auto-promotes to Shipped upon AWB input; generates direct carrier links.	VERIFIED (100%)
Alphanumeric Secure Order URLs	Customer & Admin URLs use non-guessable TSV-XXXXXXX references.	VERIFIED (100%)
Cryptographic Email Access (HMAC)	Passwordless direct-access tokens in transactional email buttons.	VERIFIED (100%)
IDOR Customer Ownership Protection	Blocks unauthorized cross-customer tracking or invoice access.	VERIFIED (100%)
De-emojified Transactional Emails	Clean, professional enterprise email cards for all 4 key lifecycle stages.	VERIFIED (100%)
Highlighted Payment Method Badges	Color-coded COD, Prepaid (Razorpay), PayPal pills across Admin, Customer & Invoice.	VERIFIED (100%)

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